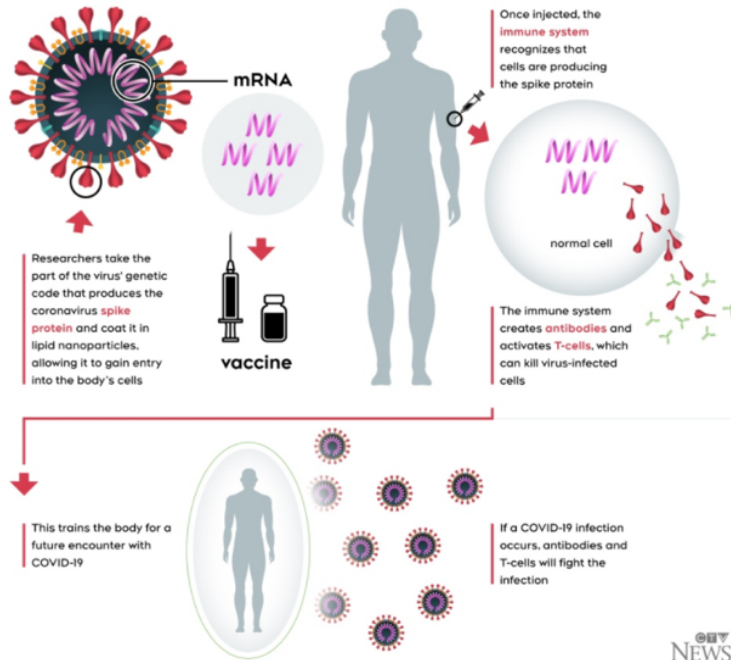


Covid-19 mRNA vaccine

Moderna, Pfizer and AstraZenica (Oxford Uni) vaccine

Evan Latef, Ph.D.
Molecular Biologist

- Moderna and Pfizer
- Both vaccines are mRNA vaccines (messenger ribonucleic acid)
 - They do NOT contain the complete virus or the complete viral genome
 - They contain a segment of the viral genome that makes the recipient's body generate an immune response as if it encountered the live virus
 - This results in the immune system creating antibodies to the virus without becoming infected first



mRNA vaccines are a new type of vaccine to protect against infectious diseases. To trigger an immune response, many vaccines put a weakened or inactivated germ into our bodies. Not mRNA vaccines.

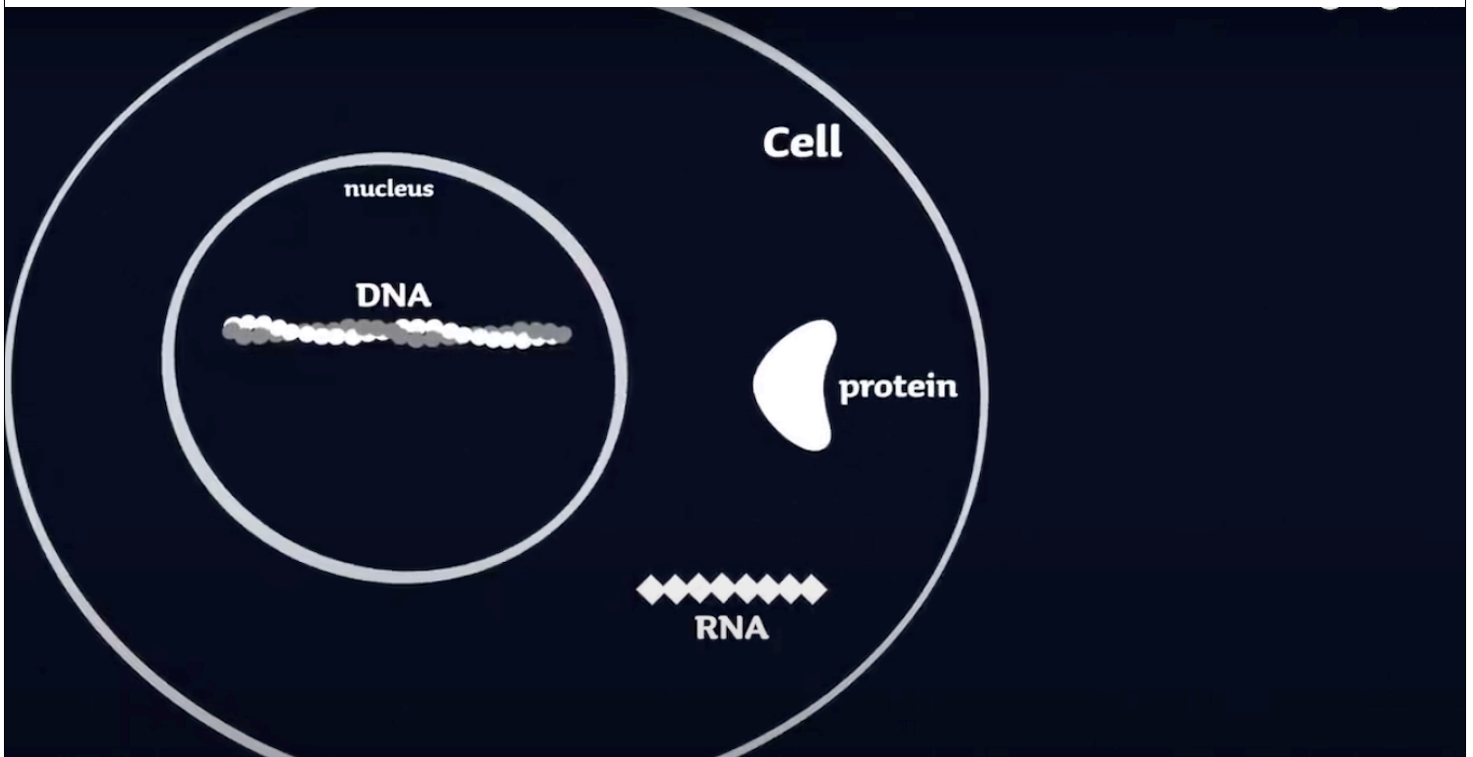
Instead, they teach our cells how to make a protein—or even just a piece of a protein—that triggers an immune response inside our bodies. That immune response, which produces antibodies, is what protects us from getting infected if the real virus enters our bodies.

Facts about Covid-19 mRNA Vaccine

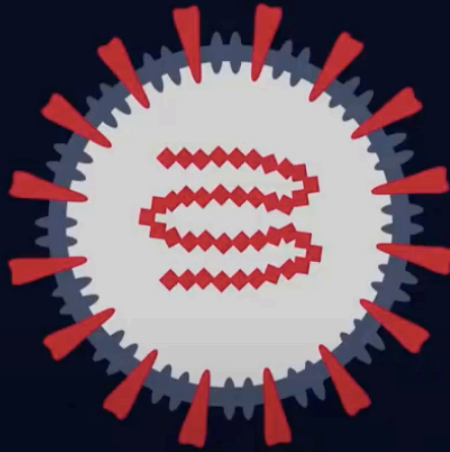
- They CANNOT give someone Covid-19.
 - mRNA vaccines do not use the live virus that causes Covid-19.
- They DO NOT affect or interact with our DNA in any way.
 - mRNA never enters the nucleus of the cell, which is where our DNA (genetic material) is kept.
 - The cell breaks down and gets rid of the mRNA soon after it is finished using the instructions.

Vaccine Side Effects

- Side effects are expected
- When the body's immune system mounts a response to a natural infection OR vaccination the result is local and/or systemic inflammation
- Local inflammation: injection site soreness, redness, swelling
- Systemic inflammation includes fever, muscle aches, headache
- The stronger the immune response, the more prominent the side effects
- The healthier the individual, the stronger the immune response

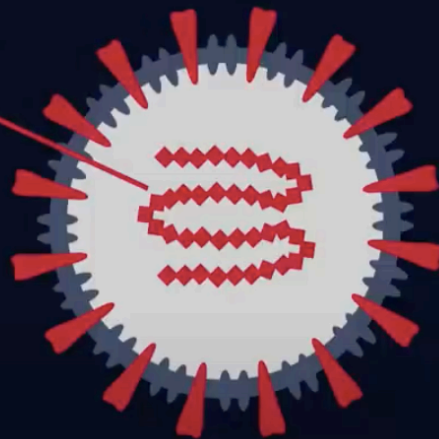


RNA Genome .



SARS-CoV-2

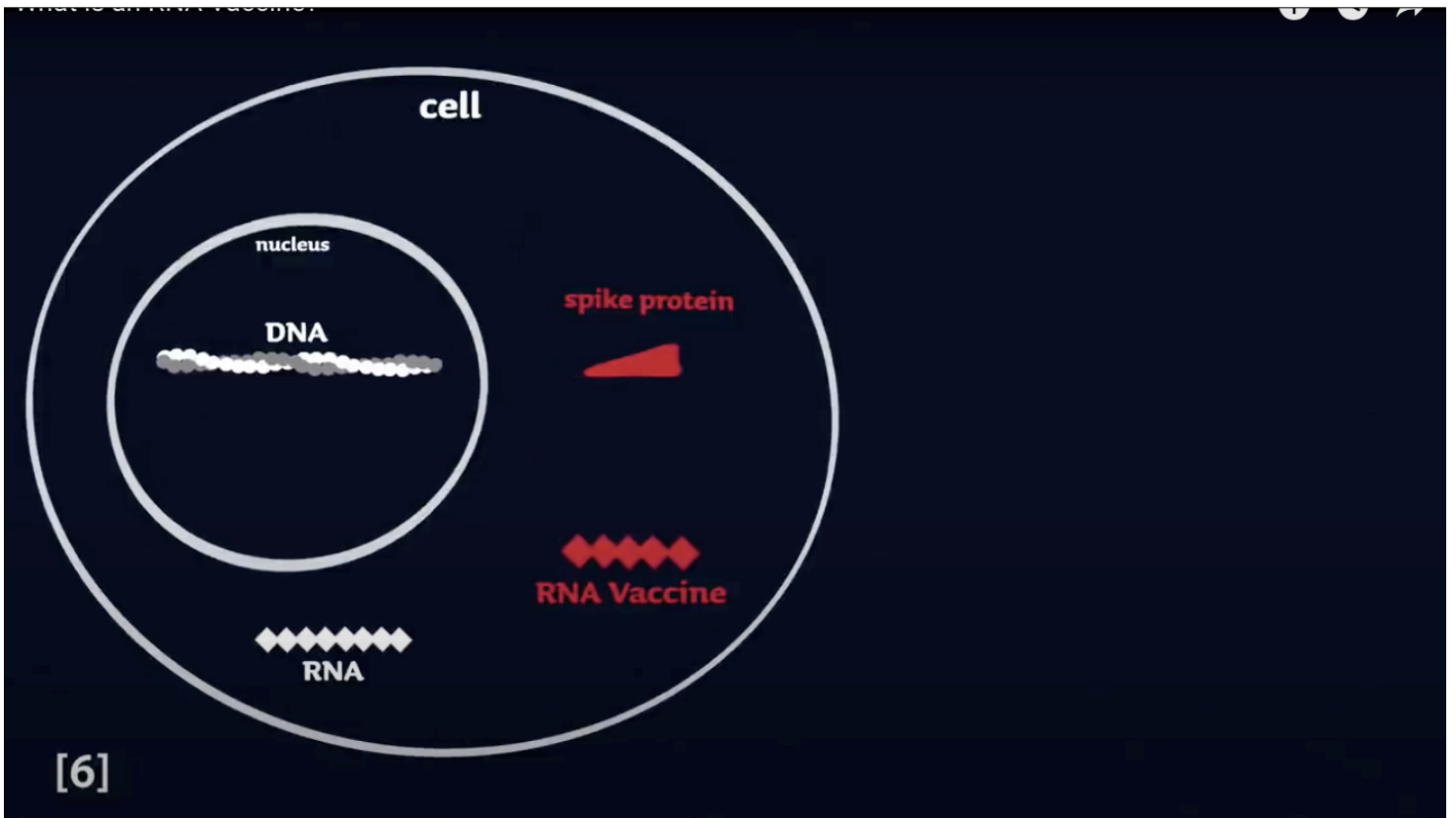
RNA Genome



Spike Protein

SARS-CoV-2

What is an RNA vaccine?



What is an RNA vaccine?

